FMGT 3510 – Finance 1
Fall 2026

Chapter 17

Working Capital Management & Short-Term Financing

Overview (1 of 2)

Working capital management involves two questions:

- What is the **right amount of operating working capital?**
- How should this amount be **financed**?

Answering these questions goes beyond finance:

- Improvements to working capital are closely related to a business' operating activities and strategy.
- Logistics, operations management, engineering, IT and other areas will be involved.
- This is because every step in the firm's operations affects working capital items:

Activity	Working Capital Accounts
Purchase raw materials	Cash, Inventory, Payables
Produce goods for sale	Cash, Inventory, Payables, Accruals
Sell goods	Cash, Receivables, Inventory
Collect receivables	Cash, Receivables
Pay payables	Cash, Payables

Overview (2 of 2)

Defining cash and short-term investments

- **Cash**: total value of short -term financial assets held to support ongoing operations i.e. cash and **equivalents** needed to run the business
- **Short term investments**: total value of short-term financial assets held for future purposes
- These definitions are consistent with our Ch. 2 definitions of Free Cash Flow and Net Operating Working Capital ("NOWC")
- They may differ from strict accounting definitions.

Difference between "**cash equivalent**" and "**short-term investment**":

normally **cash equivalent** corresponds to investment in **debt securities with less than 3 months left till maturity.**

Debt securities recorded as "**short-term investment**" normally have a time left till **maturity of more than 3 months**. Securities recorded in both categories are liquid.

Net Operating Working Capital Policies (1 of 2)

A business's NOWC policy determines the amount of NOWC used in operations.

There are three general types of policy:

- **Relaxed policy**: large amounts of cash, inventory and A/R and minimal levels of A/P and accruals, for a **given level of operations**.
- **Restricted policy**: low amounts of C/A and high levels of A/P and accruals, for a given level of operations.
- **Moderate policy**: between the two extremes

Net Operating Working Capital Policies (2 of 2)

How do you decide what policy to use? You must balance the trade-offs:

- A **relaxed policy** means that you are less likely to run out of inventory or lose sales because of tight credit policies. However, you will need to finance a large amount of NOWC which in turn will lower your return on investment ("ROI"). **Low risk, low return**.

- A **restricted policy** means higher risk of "stock outs" and lost sales but higher ROI. You will need to finance a lower amount of NOWC. **Higher risk, higher returns**.

What sort of business should have a **relaxed policy**? A **restricted policy**?

. Demand difficult to predict and/or
. Longer operating cycle and/or
. Need cushion to avoid disruption
Ex. Supermarket

. Predictable Demand and/or
. Short operating cycle and/or
. Tigh inventories
Ex. Utilities, airlines, service firms

Short Term Financing Policies (1 of 4)

- Once we have decided on the amount of NOWC we must then decide **how to finance it**.

- We must first distinguish between **temporary** NOWC and **permanent** NOWC.

- What businesses have a high level of **temporary NOWC**?
  Ski manufacturer, toy manufacturer: highly seasonal

- Who might have mostly **permanent NOWC**?
  Heavy equipment manufacturer: not seasonal

- Does the **level of permanent NOWC** stay the same?
  The level of permanent NOWC will **increase as the business grows**.

Short Term Financing Policies (2 of 4)

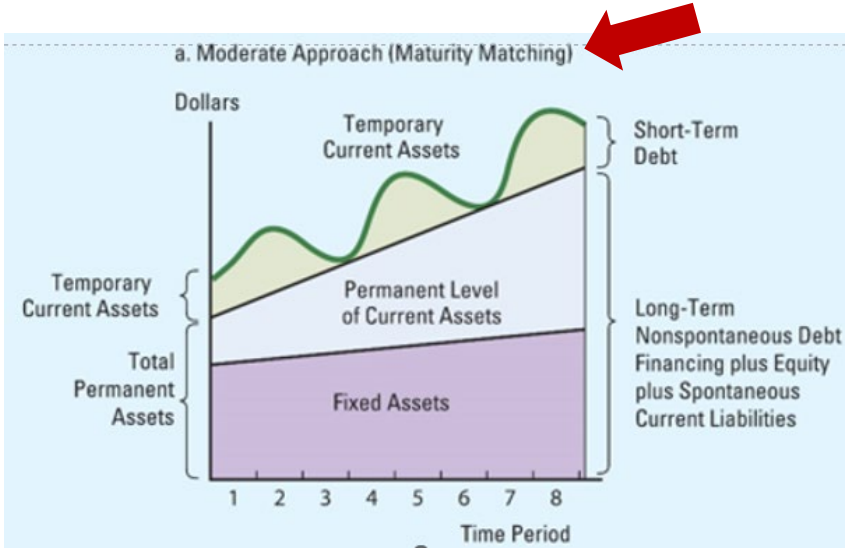
There are **three general approaches to financing permanent and temporary NOWC**:

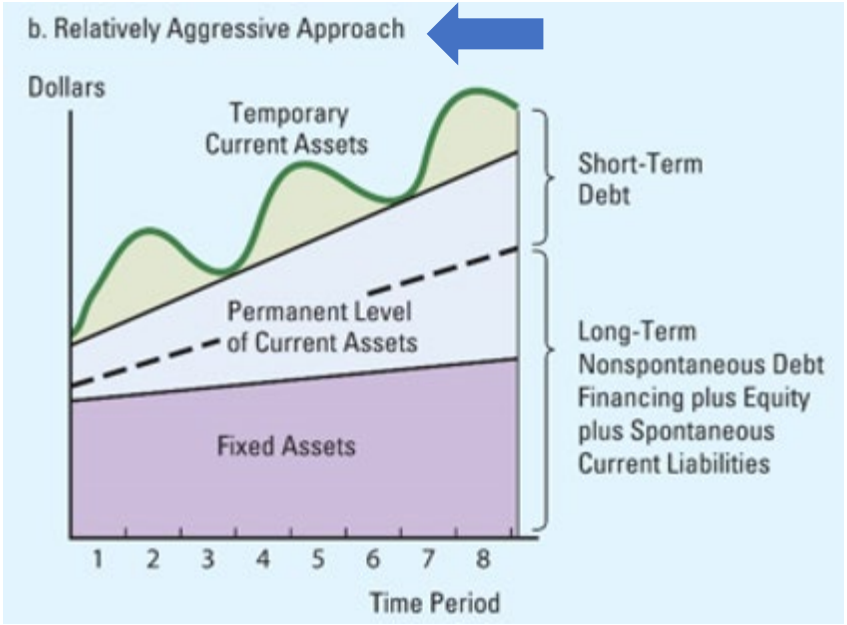
- **Maturity matching (or "self-liquidating"):**
  - Permanent NOWC financed with long term debt and/or equity
  - Temporary NOWC financed with short term debt
  - "Matching" asset term and financing term
  - Would you use a 1-year loan to finance a mine with a 30-year life?
  - Should you finance 50 days of inventory with a 5-year loan

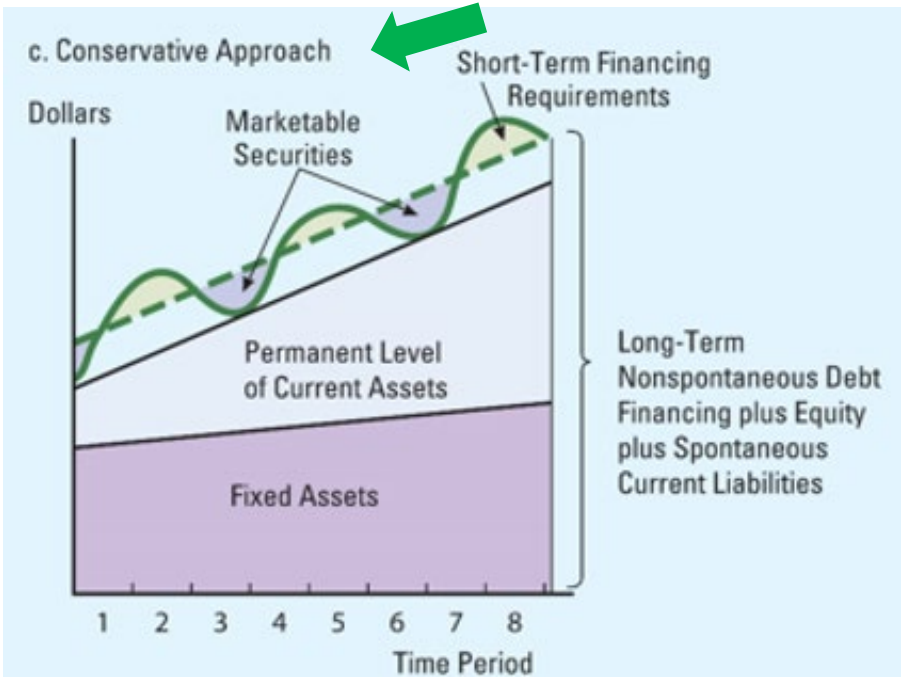
- **Aggressive policy:**
  - Use short term debt to finance some or all of your temporary NOWC and some or all of your permanent NOWC.
  - Might save interest expense because short term interest rates are usually lower than long term rates.
  - Risk that your short-term loan may not be renewed: "rollover risk"

- **Conservative policy:**
  - Use long term debt to finance all of your permanent NOWC and some or all of your temporary NOWC.
  - Rollover risk is reduced but likely face higher interest costs.
  - Need to invest excess cash when temporary NOWC is at low point.

Short Term Financing Policies (3 of 4)







Short Term Financing Policies (4 of 4)

	CONSERVATIVE			
	Jan	Feb	Mar	Apr
Temp NOWC	-	30,000	20,000	-
Perm NOWC	50,000	50,000	50,000	50,000
	50,000	80,000	70,000	50,000
Short Term Debt	-	-	-	-
Long Term Debt	80,000	80,000	80,000	80,000
Cash Surplus/(Deficit)	**30,000**	**-**	**10,000**	**30,000**

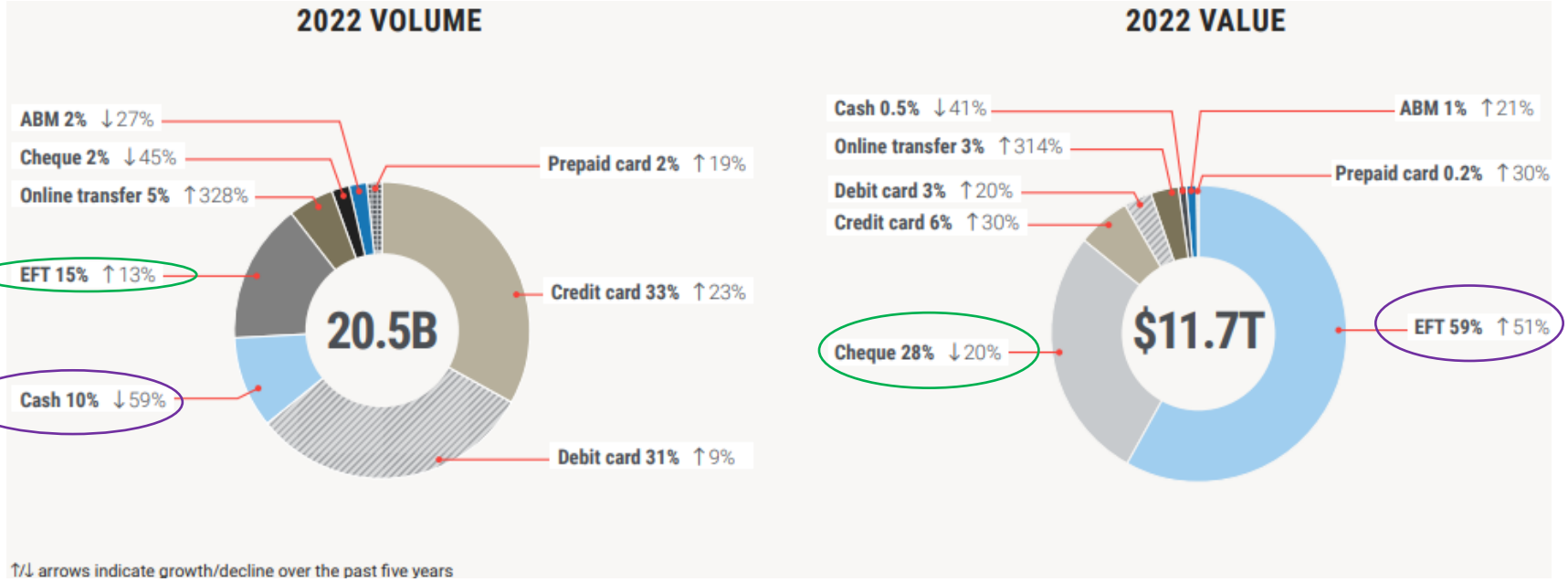
	AGGRESSIVE			
	Jan	Feb	Mar	Apr
Temp NOWC	-	30,000	20,000	-
Perm NOWC	50,000	50,000	50,000	50,000
	50,000	80,000	70,000	50,000
Short Term Debt	20,000	50,000	40,000	20,000
Long Term Debt	30,000	30,000	30,000	30,000
Cash Surplus/(Deficit)	**-**	**-**	**-**	**-**

What would a **matching** policy look like?

	Jan	Feb	Mar	Apr
ST Debt	**25,000**	**40,000**	**35,000**	**25,000**
LT Debt	**25,000**	**40,000**	**35,000**	**25,000**

Payment Methods Continue to Change (1 of 2)

The report “Canadian Payments: Methods and Trends 2023” shows how the Canadian economy continues to **evolve from physical cash and cheques and towards electronic and contactless payment methods**.



EFT: electronic funds transfers
ABM: Automated Banking Machine (Canada), also called ATM (US, and globally).

Payment Methods Continue to Change (2 of 2)

Cheque usage has declined (a 14% drop in transaction value since 2017) **but continues for large dollar amount transactions** (2% of volume but 28% of value).

Physical cash is primarily used for very low value transactions (10% of volume but only 0.5% of total value).

The COVID-19 **pandemic accelerated the trend away from physical cash**:

41% drop in value of cash transaction over the last 5 years

59% increase if EFT value over the last 5 years

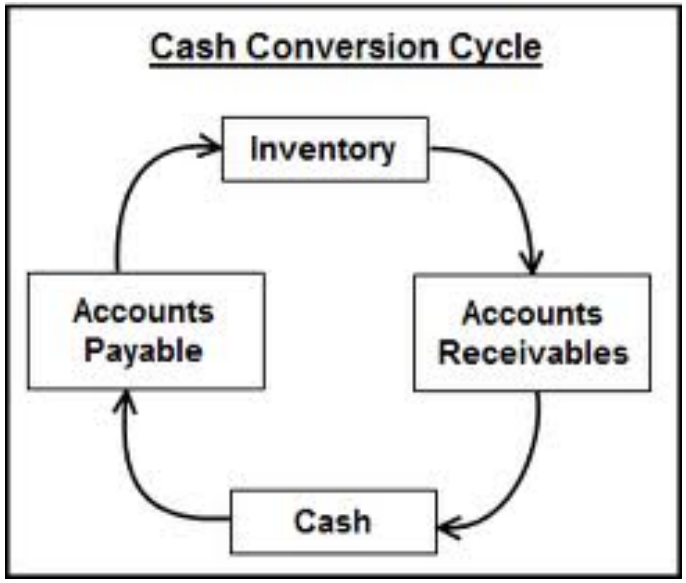
Cash Conversion Cycle (CCC) (1 of 4)

Firms follow a cycle in which they:

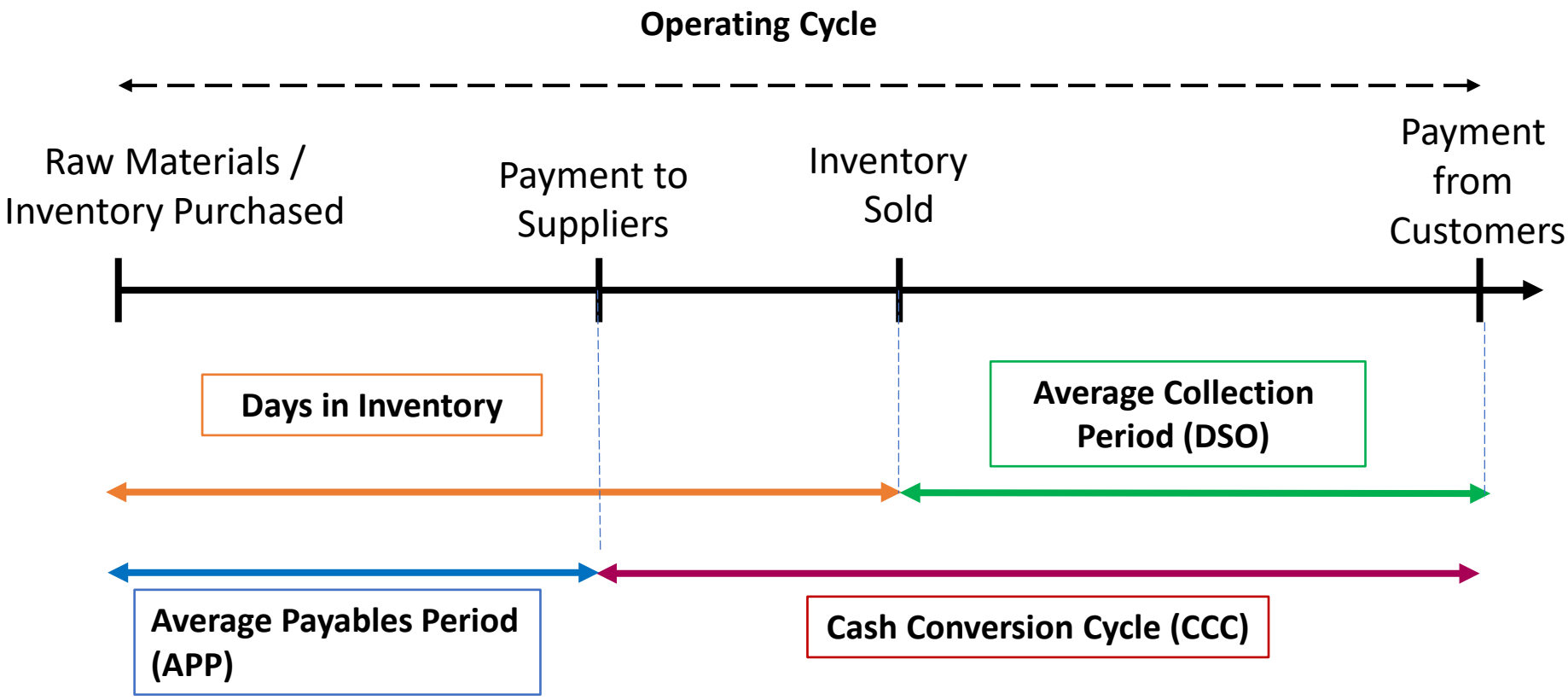
- Purchase inventory
- Sell finished goods
- Pay payables and Collect receivables

We measure this Cash Conversion Cycle ("CCC") in days:

- Inventory Conversion Period + Receivables Collection Period − Payable Deferral Period
- Days in Inventory + Days Sales Outstanding (DSO) − Average Payables Period (APP)



Cash Conversion Cycle (CCC) (2 of 4)



Cash Conversion Cycle (CCC) (3 of 4)

Example 1. Calculate the **Cash Conversion Cycle** using the following balances.

- Accounts Receivable: $800,000
- Inventory: $2,500,000
- Accounts Payable: $910,000

Sales (all credit) were $10,000,000 and cost of goods sold was $8,000,000.

Determine the CCC.

$$\text{ICP} = \frac{2{,}500{,}000}{\left(\frac{8{,}000{,}000}{365}\right)} = 114 \text{ days} \qquad \text{DSO} = \frac{800{,}000}{\left(\frac{10{,}000{,}000}{365}\right)} = 29 \text{ days}$$

$$\text{PDP} = \frac{910{,}000}{\left(\frac{8{,}000{,}000}{365}\right)} = 42 \text{ days } (41.52)$$

$$\text{CCC} = \mathbf{114 + 29 - 42 = 101 \text{ days}}$$

Cash Conversion Cycle (CCC) (4 of 4)

If the **ICP** is **cut** to **110** days and the **DSO** to **25** days and the **PDP** increases to **45** days what is the **new CCC**? By how much **is free cash flow increased** in the year of these changes? In the next year, if **sales increase by 10%?**

A/R drops by 4 days: **(10,000,000/365) x 4=**	**$109,589 increase** in cash flow
Inventory drops by 4 days: **(8,000,000/365) x 4 =**	**$87,671 increase** in cash flow
A/P increased by 3 days: **(8,000,000/365) x 3 =**	**$65,753 increase** in cash flow
Total change	**$263,013 increase** in cash flow

10% growth in sales, with the improved CCC, will **generate** another cash improvement of: **(10%)($263,013) = $26,301**

Review of Cash budgeting

The cash budget is the primary tool for short-term financial planning. It helps to identify short-term financial needs (i.e. borrowing) and opportunities.

Cash inflows:

Collection of cash sales and accounts receivable

Sale of capital assets

Issuance of **additional** debt and equity

Cash outflows

Payment of accounts payable

Payment of wages, taxes and other cash expenses

Capital expenditures ("capex")

Long-term financing expenditures

- Cash dividends
- Interest
- Repayment of debt principal.

Cash Budget vs. Income Statement and Free Cash Flow (1 of 2)

There are **significant differences between a cash budget and the income statement**

- The **income statement records sales**, not the **collection of cash from those sales**.
- It **accrues expenses** based on the matching principle and **not based on actual cash payments**. For example, income tax expense is matched to income and not to the actual cash payment of taxes, which occurs later.
- The **income statement includes depreciation**, which is not a use of cash. It also does not show cash expenditures on long-lived or capital assets.
- It **expenses materials purchased when the associated goods are sold**, not when the materials are paid for.

Cash Budget vs. Income Statement and Free Cash Flow (2 of 2)

The **cash budget differs from free cash flow** in part because it **includes payments to investors (e.g. dividends, interest, etc.)**

- The calculation of **free cash flow**, on the other hand, shows how much is **available to be paid to investors** (or is required from them)
- As a result, **free cash flow** projections can be used to **value** a business by calculating the present value of the cash flows available to investors.

Cash Budget Example (see excel spreadsheet)

We will use the following assumptions and the handout template to construct a cash budget.

Accounts receivable

- 20% is collected in cash in the same month that the sale occurs
- A 2% discount is granted if the account is paid in the same month as the sale
- 70% is collected in the month after the sale
- 10% is collected two months after the sale

Accounts payable and purchases

- Purchases are made monthly in an amount equal to 70% of the next month's forecast sales.
- Accounts payable are paid in full in cash the month after the purchase is made. $400,000 will be spent on new equipment in October
- Other expenses are paid in cash as shown on the handout.
- The firm's policy is to maintain a cash balance of at least $10,000

Short term financing – overview

Advantages of short-term financing

- Faster/easier to obtain.
- Appropriate for temporary NOWC needs caused by seasonality.
- Not "locked in" to an amount or an interest rate for a long time.
- Fewer covenants then for long term debt.
- Usually cheaper.

Disadvantages of short-term financing

- Volatile interest cost as rates change.
- No long-term commitment by lenders-face "roll over" risk if loan not renewed.

*What are the **different types of short-term financing**? How do we choose between them?*

Short term financing – Accounts Payable (trade credit) (1 of 3)

Most common form of short-term financing. It is usually easy to obtain, and it is a source of **spontaneous financing** (expands and contracts along with sales).

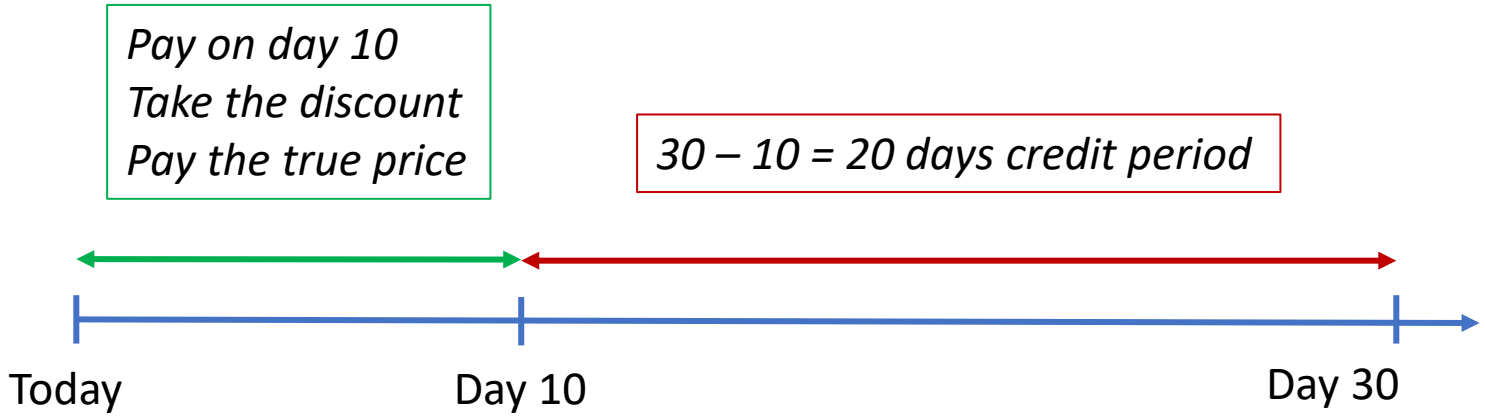
The terms of credit usually provide for a **discount for early payment**

Ex. 2% if paid in the first 10 days and payable in full after 30 days or "2/10, net 30"

The cost of trade credit is the **cost of not taking the discount** offered.

A supplier might offer us a **2% discount if we pay in 10 days** instead of 30 days.

If we don't take the discount, then we are effectively **paying 2% to borrow money for an extra 20 days i.e. 30 – 10 = 20 days.**



Short term financing – Accounts Payable (trade credit) (2 of 3)

The **annual percent rate (APR)** of not taking the discount (Nominal annual cost of trade credit) is:

$$\text{APR} = \frac{\text{Discount Percent}}{100\% - \text{Discount Percent}} \times \frac{365}{\text{Days Credit} - \text{Discount Period}}$$

$$\text{APR} = \frac{2\%}{100\% - 2\%} \times \frac{365}{30 - 10} = 0.372\ (37.2\%)$$

The **effective annual rate takes into account** the compounding periods:

$$\text{EAR} = \left[1 + \frac{\text{Discount Percent}}{100 - \text{Discount Percent}}\right]^{\frac{365}{\text{Days Credit} - \text{Discount Period}}} - 1$$

$$\text{EAR} = \left[1 + \frac{2}{100 - 2}\right]^{\frac{365}{30-10}} - 1 = 0.446\ (44.6\%)$$

Short term financing – Accounts Payable (trade credit) (3 of 3)

Example 3. You purchase goods on terms of 2/10, net 30.

Cost of financing if you pay on day 35

$$\text{APR} = \frac{2}{100-2} \times \frac{365}{35-10} = 0.298\ (29.8\%)$$

$$\text{EAR} = \left(1 + \frac{2}{98}\right)^{\left(\frac{365}{35-10}\right)} - 1 = 0.3431\ \mathbf{(34.31\%)}$$

Cost of financing if you pay on day 45

$$\text{APR} = \frac{2}{100-2} \times \frac{365}{45-10} = \ (\%)$$

$$\text{EAR} = \left(1 + \frac{2}{98}\right)^{\left(\frac{365}{45-10}\right)} - 1 = 0.2345\ \mathbf{(23.45\%)}$$

Short term financing – Short Term Bank Loan (1 of 2)

Short-term bank loans are the main source of **non-spontaneous financing**. The business must first apply for a loan and then negotiate a loan agreement with the bank.

- Loan maturity: 1 year or less.
- Interest rate and conditions (ex. length, covenants, collateral) depend on the firm's default risk
- To assess the firm's default risk, the bank will analyze its past and forecasted financial statements (*pro forma statements*).
- **What ratios will the bank focus on? CR, QR, TIE, DR, D-to-E, EBITDA coverage …**

The interest rate will usually be **floating rate** based on the **bank prime rate** or a **market based floating rate**.

The use of LIBOR, the market-based floating interest rate, has ended in 2021. It is being replaced by **rates calculated and published by each country's central bank** based in real transactions (verifiable). For example:

- **CORRA** for Canada (**Canadian Overnight Repo Rate Average**)
- **SOFR** for USA (**Secured Overnight Financing Rate**)

Short-term bank loans can be interest only or amortized.

Short term financing – Short Term Bank Loan (2 of 2)

The firm will likely have to provide **collateral** security. Examples are:

- A/R
- Inventory
- Personal guarantee

The loan agreement will include **covenants**

Covenants are promises made by the borrower to do certain things (**positive** covenants) and/or not do certain other things (**negative** covenants)

*Ex: **positive** covenant: CR should be greater than 1.8 x*

*Ex. **negative** covenant: Dividend payout ratio can Not exceed 30%*

Short term financing – Line of credit

The **typical loan is a one-year line of credit**. The loan states the maximum amount that can be borrowed. The firm can borrow any amount up to this maximum during the term of the loan.

Main features of a Line of credit

- Normally 1 year loan designed to **finance seasonal working capital requirements**.
- Interest are usually stated as a specified spread (*premium*) over some kind of **market reference rate (ex. Prime rate + 0.5%).** This spread reflects the firm's default risk.
- The bank also charges a fee (ex. 0.5% of the amount used or, sometimes, of the amount unused). **No fee if the line of credit is uncommitted.**
- The line of credit can be **committed** or **uncommitted**
  - **Committed**: the bank can not refuse to lend over the duration of the contract.
  - **Uncommitted**: the bank can refuse to lend, for example if the firm's default risk increase.
- **Revolving line of credit**: normally for 3 to 4 years. Provides a more stable source of financing. Lower rollover risk. Revolving line of credit are riskier for the bank, and therefore will come with specific covenants (ex. Maintain QR of at least …)

Short term financing – Money Market

Large firms with excellent credit records can obtain financing **directly from the money markets by issuing short-term debt securities.**

- **Commercial paper**: short-term notes with maturities ranging from 30 to 365 days, offered in denominations of $100 thousand and up. Usually backed by a bank line of credit. Commercial paper is usually less costly than bank loans.
- **Bankers' acceptances**: variant of commercial paper where for a fee, a bank guarantees the paper's principal and interest.

Disadvantage of money market financing: the risk that the market may temporarily **dry up** (as it in during the 2008 credit crisis) when it comes time to **roll over** the paper **and interest rates volatility.**

Calculating Financing costs (1 of 4)

The effective cost on a loan depends on **how often the interest is paid** (the more frequently, the higher the rate).

The general approach is to **determine the loan's cash flows** and **then calculate the IRR, making sure to convert it to the EAR.**

- Treat a loan as you would any other financial asset and use time value of money techniques to calculate its rate of return.
- **Consider any fees and expenses as well as the timing of the loan's principal repayment and interest payments.**
- **Compensating balance.**

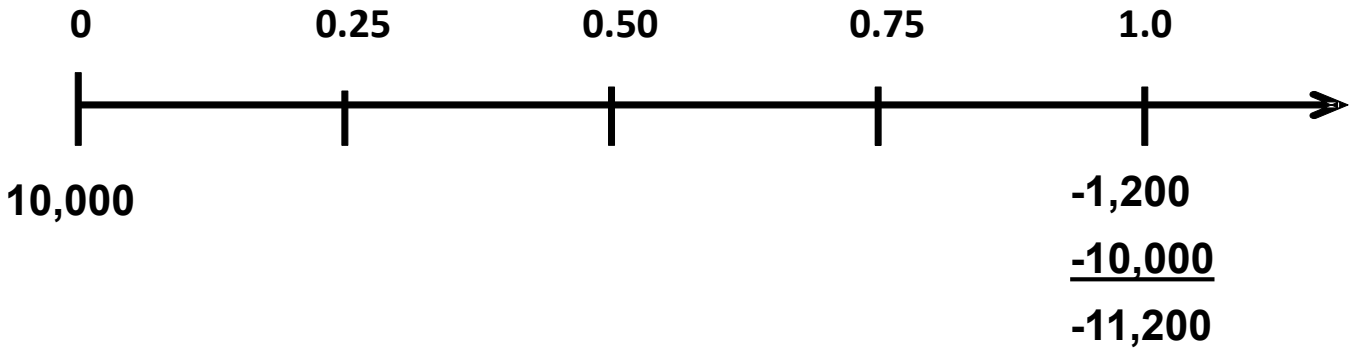
Remember that the pre-tax return to the lender will be the same as the pre-tax cost to the borrower

Calculating Financing costs (2 of 4)

Example 4 – ADD ON BASIS

Assuming Interest Paid Annually

This is the simplest case: you borrow $10,000 for one year at an APR of 12%.

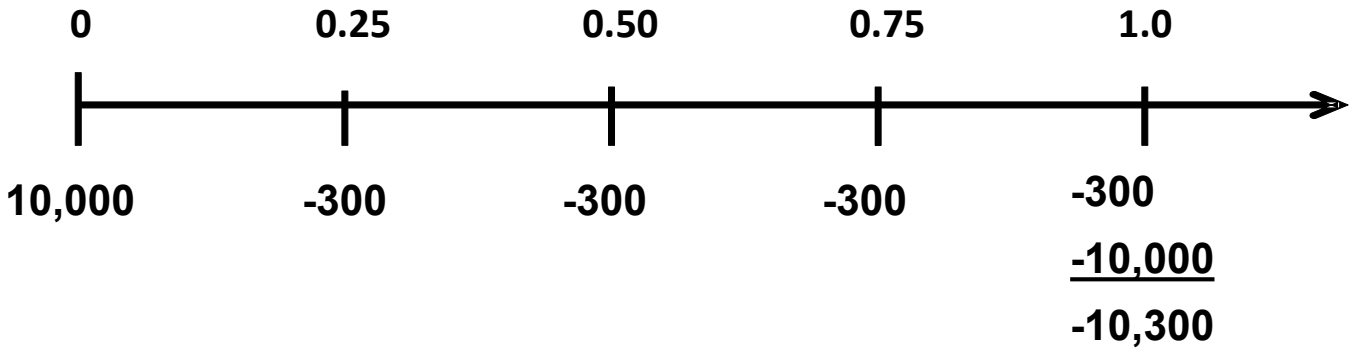


P/Y = 1 C/Y = 1	N	i/Y	PV	PMT	FV
Input	1		10,000	0	- 11,200
CPT		**12**			

Calculating Financing costs (3 of 4)

Example 5 – ADD ON BASIS

What if APR is 12%, paid quarterly? Let's assume that each quarter is the same length.

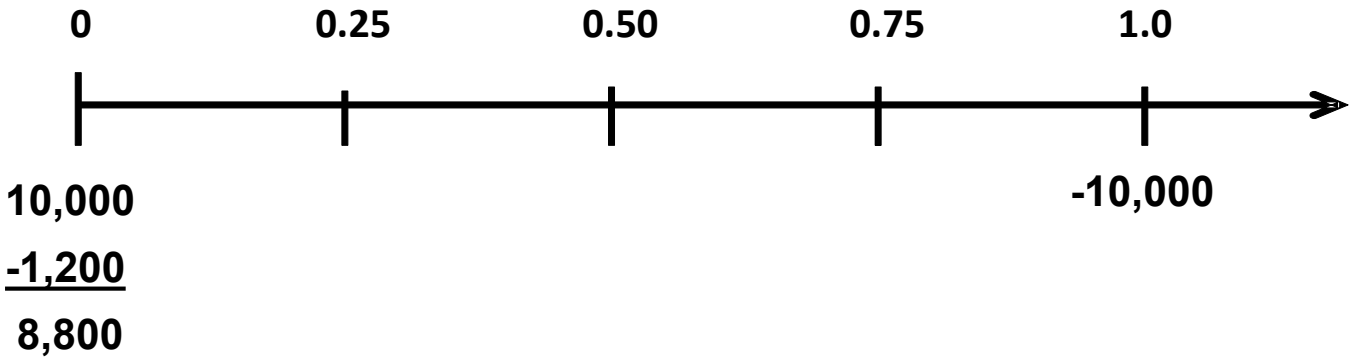


P/Y = 4 C/Y = 1	N	i/Y	PV	PMT	FV
Input	4		10,000	-300	-10,000
CPT		**12.55**			

Calculating Financing costs (4 of 4)

Example 6 – DISCOUNT BASIS (*method used for Commercial Paper and T-bills*)

What if interest rate is 12% but paid on a discount basis? This means that the interest amount ($1,200) is paid at the beginning of the loan period.



P/Y = 1 C/Y = 1	N	i/Y	PV	PMT	FV
Input	1		8,800	0	- 10,000
CPT		**13.64**			

Secured Short Term Financing (1 of 4)

Secured loans – most operating loans are secured with assets of the borrower

Pledging Accounts Receivable

- Accounts receivable are pledged to the lender as security.
- **Borrower is responsible for collecting the receivables**.
- Lender may review all invoices, if the number is not large, and determine which to take as security.
- The size of the loan is a function of the **margin percentage** (usually 75% of current receivables).
- The **lender retains recourse against the borrower**: if the A/R becomes a "bad debt" the borrower must still pay back the loan.

Secured Short Term Financing (2 of 4)

Secured loans – most operating loans are secured with assets of the borrower

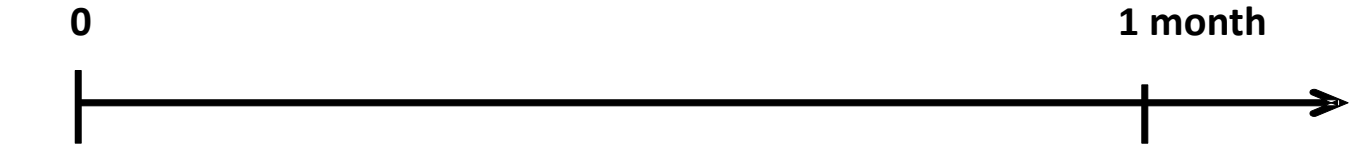
Factoring receivables

- **Borrower actual sells the receivables to a factor** who is then responsible for collecting them.
- Factor will charge a fee for collecting the receivables and assuming the credit risk. The factor will also charge interest on funds advanced to the seller before collection.
- The seller may realize cost savings by transferring the credit and collection functions to the factor.
- **Normally, with recourse**: the firm (**borrower**) remains responsible for the AR. If the client fails to pay, the firm will have to pay the factoring company.

Secured Short Term Financing (3 of 4)

Example 7 – Cost of factoring

The factor charges a fee equal to 2.5% of A/R sold to it, plus interest of 9% APR and a deposit equal to 5% of A/R. What is the EAR of factoring a 1-month A/R of $10,000?



10,000		
-500	deposit	
-75	interest	= 10,000(0.09/12)
-250	fee =	0.025(10,000)
9,175		

-10,000		
500	deposit	
-9,500		

P/Y = 1 C/Y = 1	N	i/Y	PV	PMT	FV
Input	1		9,175	0	- 9,500
CPT		**3.542**			

Monthly rate

$$\mathbf{EAR} = (\mathbf{1} + \mathbf{0.03542})^{\mathbf{12}} - \mathbf{1} = \mathbf{0.5184}\ \boldsymbol{or}\ \mathbf{51.84\%}$$

Secured Short Term Financing (4 of 4)

Inventory financing

- Inventory is assigned to lender as security based on a predetermined margin percentage (50%) (usually a lower percentage than the one used for receivable).
- Form of charge:
  - **Blanket lien (covering all inventory)**
  - **Trust receipts** (covering specific, high value items such as automobiles)
  - **Warehouse receipt financing**: goods are held in either a **third-party public warehouse** or at a **field warehouse** at the borrower's premises but segregated.